Introduction to Artificial Intelligence and Machine Learning (CSN11601)

Assignment-6

Name : Aayansh Niranjan Date : 13/10/2025
Reg_No.: 20253004

Q1.Apply different pre-processing steps written below for the Titanic dataset.

a) Write the Code to load the csv file.
b) Write the code to find missing values in each column.
c) Write the code to drop columns from the data.
d) Write the code to convert 'male' to 1 and 'female' to 0 5.
e) Write the code to fill missing values in 'Age' column.
f) Write the code to split the Input and Output columns.
g) Write the code the data in train-test set.

(Dataset Link: https://github.com/datasciencedojo/datasets/blob/master/titanic.csv)

A1. CODE:

```
import pandas as pd
from sklearn.model_selection import train_test_split
df=pd.read_csv('titanic.csv')
print(df.head())

missing_values=df.isnull().sum()
print("\nMissing values in each column:")
print(missing_values)
df=df.drop(['Name', 'Ticket', 'Cabin'], axis=1)
print("\nData after dropping unnecessary columns:")
print(df.head())
df['Sex']=df['Sex'].map({'male':1,'female':0})
print("\nData after converting 'male' to 1 and 'female' to 0:")
print(df.head())
df['Age']=df['Age'].fillna(df['Age'].median())
print("\nData after filling missing values in 'Age' column:")
print(df.head())
X=df.drop('Survived',axis=1)
y=df['Survived']

print("\nInput (X) and Output (y) columns split:")
print("Input (X):")
print(X.head())
print("Output (y):")
print(y.head())
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print("\nData split into train and test sets:")
print(f"Training Data (X_train):\n{X_train.head()}")
print(f"Testing Data (X_test):\n{X_test.head()}")
```

OUTPUT:

```
(base) user@wipro-desktop:~/Aayansh_20253004/Assignment-6$ python p1.py
   PassengerId  Survived  Pclass                                               Name     Sex   Age  SibSp  Parch            Ticket     Fare Cabin Embarked
0            1         0       3                            Braund, Mr. Owen Harris    male  22.0      1      0         A/5 21171   7.2500   NaN        S
1            2         1       1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1      0          PC 17599  71.2833   C85        C
2            3         1       3                             Heikkinen, Miss. Laina  female  26.0      0      0  STON/O2. 3101282   7.9250   NaN        S
3            4         1       1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1      0            113803  53.1000  C123        S
4            5         0       3                           Allen, Mr. William Henry    male  35.0      0      0            373450   8.0500   NaN        S

Missing values in each column:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64

Data after dropping unnecessary columns:
   PassengerId  Survived  Pclass     Sex   Age  SibSp  Parch     Fare Embarked
0            1         0       3    male  22.0      1      0   7.2500        S
1            2         1       1  female  38.0      1      0  71.2833        C
2            3         1       3  female  26.0      0      0   7.9250        S
3            4         1       1  female  35.0      1      0  53.1000        S
4            5         0       3    male  35.0      0      0   8.0500        S

Data after converting 'male' to 1 and 'female' to 0:
   PassengerId  Survived  Pclass  Sex   Age  SibSp  Parch     Fare Embarked
0            1         0       3    1  22.0      1      0   7.2500        S
1            2         1       1    0  38.0      1      0  71.2833        C
2            3         1       3    0  26.0      0      0   7.9250        S
3            4         1       1    0  35.0      1      0  53.1000        S
4            5         0       3    1  35.0      0      0   8.0500        S

Data after filling missing values in 'Age' column:
   PassengerId  Survived  Pclass  Sex   Age  SibSp  Parch     Fare Embarked
0            1         0       3    1  22.0      1      0   7.2500        S
1            2         1       1    0  38.0      1      0  71.2833        C
2            3         1       3    0  26.0      0      0   7.9250        S
3            4         1       1    0  35.0      1      0  53.1000        S
4            5         0       3    1  35.0      0      0   8.0500        S
Input (X) and Output (y) columns split:
Input (X):
   PassengerId  Pclass  Sex   Age  SibSp  Parch     Fare Embarked
0            1       3    1  22.0      1      0   7.2500        S
1            2       1    0  38.0      1      0  71.2833        C
2            3       3    0  26.0      0      0   7.9250        S
3            4       1    0  35.0      1      0  53.1000        S
4            5       3    1  35.0      0      0   8.0500        S
Output (y):
0    0
1    1
2    1
3    1
4    0
Name: Survived, dtype: int64

Data split into train and test sets:
Training Data (X_train):
     PassengerId  Pclass  Sex   Age  SibSp  Parch     Fare Embarked
331          332       1    1  45.5      0      0  28.5000        S
733          734       2    1  23.0      0      0  13.0000        S
382          383       3    1  32.0      0      0   7.9250        S
704          705       3    1  26.0      1      0   7.8542        S
813          814       3    0   6.0      4      2  31.2750        S
Testing Data (X_test):
     PassengerId  Pclass  Sex   Age  SibSp  Parch     Fare Embarked
709          710       3    1  28.0      1      1  15.2458        C
439          440       2    1  31.0      0      0  10.5000        S
840          841       3    1  20.0      0      0   7.9250        S
720          721       2    0   6.0      0      1  33.0000        S
39            40       3    0  14.0      1      0  11.2417        C
(base) user@wipro-desktop:~/Aayansh_20253004/Assignment-6$
```